Scalability & Future Plan

Date	03 October 2026
Team ID	6ab4fdb0a238dc999eb66571
Project Name	ComicCraft
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Local SQLite storage	Limited concurrent writes for large deployments	Medium
2	Dependence on external AI API	Availability and latency depend on provider	High
3	Text-first output in MVP	Does not directly produce finished comic artwork	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single/local server	Deploy behind Nginx with multiple Flask workers/containers.
2	Data Storage	SQLite	Migrate to PostgreSQL with indexed project tables.
3	Performance	Synchronous AI calls	Use async jobs, caching and queue-based processing.
4	Security	Environment-based API key	Add centralized secrets management, rate limiting and audit logs.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact

Phase 2			
Phase 3	Image-generation integration and character consistency	Q2 2027	Move from storyboard to visual comic creation
Phase 4	Collaboration, templates and analytics	Q3 2027	Support team workflows and creator reuse